

Linear algebra: Exercise MATLAB session

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1. Write a MATLAB code that prompts the user to enter a scalar as the dimension of a vector space and displays the elementary, zeros and ones vectors in this space.
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2. Write a MATLAB code that prompts the user to enter a vector and a scalar, then displays the vector, the dimension of vector space containing the vector, the length (norm) of the vector, and the scalar multiplication of the scalar with the vector.
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3. Write a MATLAB code that prompts the user to enter two vectors, then displays the addition and the inner product of them, and the angle value between them.
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4. Write a MATLAB code that prompts the user to enter a vector whose elements are the entries of a diagonal matrix, then displays the matrix and the zeros and identity matrices with the size of diagonal matrix.
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5. Write a MATLAB code that prompts the user to enter a matrix, then displays the matrix and the size, transpose, and trace of matrix.
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6. Write a MATLAB code that prompts the user to enter a matrix, a vector, and a scalar, then displays the matrix, the vector, and the scalar-matrix and vector-matrix multiplications.
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7. Write a MATLAB code that prompts the user to enter two matrices, then displays addition of matrices and the matrix-matrix multiplication of them.
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